Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 30: Leaning on Every Earlier Case

Recursive Sequences and Strong Induction

Chartrand & Zhang, Sections 4.3 and 4.4, pp. 128–142

Name: _____ Date: _____

Where we are

Ordinary induction lets you assume $P(k)$ to prove $P(k+1)$. Sometimes one step back is not enough — the next case depends on the case two steps back, or on some earlier case you cannot predict. Today: a version of induction that lets you assume *all* of them.
This is also the mathematics behind every recursive function you write.

By the end of today you should be able to

1. Compute terms of a recursively defined sequence and conjecture a formula.
2. Prove a formula for a recursive sequence by induction.
3. Recognise when strong induction is needed, and use several base cases.

1. Recursively defined sequences

Recursive definition

A sequence is **defined recursively** by giving one or more **initial values** and a **recurrence relation** expressing each later term using earlier ones.

Example. $a_1 = 3$ and $a_n = 2a_{n-1} + 1$ for $n \geq 2$.

$a_2 = \underline{7}$, $a_3 = \underline{15}$, $a_4 = \underline{31}$. Conjecture: $a_n = \underline{2^{n+1} - 1}$.

Result 4.29. *Proof by ordinary induction.* Base: $a_1 = 3 = 2^2 - 1$. Assume $a_k = 2^{k+1} - 1$. Then

$$a_{k+1} = 2a_k + 1 = 2\big(2^{k+1} - 1\big) + 1 = \underline{2^{k+2} - 1}.$$

By induction the formula holds for all n. ■

Each term depends only on the one before, so assuming $P(k)$ was enough.

2. When one step back is not enough

The Strong Principle of Mathematical Induction

The statement $\forall n \in \mathbb{N},\ P(n)$ is true if

(1) $P(1)$ is true, and

(2) $\forall k \in \mathbb{N},\ P(1) \wedge P(2) \wedge \cdots \wedge P(k) \Rightarrow P(k+1)$ is true.

The only difference: in the inductive step you may assume every earlier case, not just the last one.

Result 4.30. $a_1 = 2$, $a_2 = 3$, and $a_n = 2a_{n-1} - a_{n-2}$ for $n \geq 3$. Then $a_n = n + 1$.

Proof. We use strong induction. Base: $a_1 = 2 = 1 + 1$.

Assume $a_i = i + 1$ for every i with $1 \leq i \leq k$. We show $a_{k+1} = k + 2$.

If $k = 1$: $a_2 = 3 = 1 + 2$, given directly.

If $k \geq 2$: then $k + 1 \geq 3$, so the recurrence applies:

$$a_{k+1} = 2a_k - a_{k-1} = 2(k+1) - \big((k-1)+1\big) = k + 2.$$

By strong induction, $a_n = n + 1$ for all n. ■

Why $k = 1$ needed separate treatment

The recurrence only holds for $n \geq 3$. When $k = 1$ we would need $a_2 = 2a_1 - a_0$, and there is no a_0. Chartrand flags exactly this.

Check that the recurrence actually applies before you use it. With two initial values, the first step past them usually needs its own case — or its own base step.

3. Several base cases at once

Claim

Every amount of postage of 12 cents or more can be made using only 4-cent and 5-cent stamps.

Base cases. Fill in:

$$12 = 4 + 4 + 4, \quad 13 = 4 + 4 + 5, \quad 14 = 4 + 5 + 5, \quad 15 = 5 + 5 + 5.$$

Inductive step. Let $k \geq 15$ and assume every amount from 12 to k can be made. We show $k + 1$ can. Since $k + 1 \geq 16$, we have $(k + 1) - 4 \geq 12$, so by the inductive hypothesis $k - 3$ can be made. Add one 4-cent stamp. ■

Why four base cases

The inductive step reaches back **four** places, to $k - 3$. So it only works once four consecutive amounts are known. Fewer base cases leave a gap the chain cannot cross.

And the threshold is real: 11 cents *cannot* be made from 4s and 5s. Test it.

4. Well-ordering, and why these are all the same

The Well-Ordering Principle

Every nonempty set of positive integers has a **least element**.

This supports a style of proof: suppose a claim fails, let n be the *smallest* counterexample, and derive a contradiction by finding an even smaller one. That is proof by contradiction from Lecture 27, aimed at a minimal failure.

Three formats, one idea

Ordinary induction, strong induction, and well-ordering are **logically equivalent** — any proof in one format can be rewritten mechanically in the others (*MCS* Section 5.3).
So why keep all three? Because they signal different things to a reader: ordinary induction says each case follows from the previous one; strong induction says it may need earlier ones; well-ordering says "look at the smallest thing that goes wrong." Choose the one that tells your reader the truth about the argument.

This is recursion

A recursive function is correct exactly when its base cases are right and each call is correct assuming the smaller calls are. That is strong induction, word for word. A missing base case in a proof is the same bug as a missing base case in code — and it fails the same way.

Exit ticket

1. $a_1 = 1$, $a_2 = 4$, $a_n = 2a_{n-1} - a_{n-2} + 2$. Compute a_3, a_4, a_5 and conjecture a formula.

2. When is strong induction needed rather than ordinary induction?

Before Monday

Read the Chapter 4 Highlights, pp. 143–145.

Self-check (not collected). Section 4.4, Exercises 1, 3, and 5 — odd-numbered, answers in the back.

Coming up

Homework 10 is due tonight, Friday, October 30, 11:59 p.m.
Quiz 9 is Monday, November 2 — Sections 4.1 and 4.2.
Homework 11 covers Chapter 4 and is due Friday, November 6. **Counting begins Monday**, with Chapter 8.

Questions I still have